

## What is TensorFlow?

TensorFlow is an open-source Python library developed by Google for Machine Learning (ML) and Deep Learning (DL). It is used to build, train, and deploy Artificial Intelligence (AI) models.

Developed by: Google Brain Team

Released: 2015

Website: <https://www.tensorflow.org>

## Features of TensorFlow

- Open-source and free Supports Machine Learning and Deep Learning
- High performance and fast computation
- CPU, GPU, and TPU support
- Automatic differentiation
- Scalable for small and large datasets
- Cross-platform support (Windows, Linux, macOS)
- Easy deployment on mobile, web, and cloud
- Integrates with NumPy, Pandas, and Matplotlib
- Provides Keras API for easy neural network development

## Advantages of TensorFlow

- Easy to build deep learning models
- Supports distributed computing
- Excellent documentation and community support
- High performance with GPU and TPU
- Suitable for research and production
- Supports image, text, speech, and video processing

## Disadvantages of TensorFlow

- Difficult for beginners
- Requires more memory
- Deep learning models can take longer to train

- Debugging can be complex for large models

## Installation

```
pip install tensorflow
```

## Import TensorFlow

```
import tensorflow as tf
```

## Check Version

```
import tensorflow as tf print(tf.__version__)
```

## Tensor

A Tensor is the basic data structure in TensorFlow. It is a multi-dimensional array used to store data.

## Example

```
import tensorflow as tf a = tf.constant([1,2,3]) print(a)
```

## Data Types

TensorFlow supports many data types.

- int32
- int64
- float32
- float64
- bool
- string

## Example

```
tf.constant([1,2,3], dtype=tf.float32)
```

## Variables

Variables store values that can be changed during model training.

```
v = tf.Variable([1,2,3]) print(v)
```

## Basic Operations

### Addition

```
tf.add(10,20)
```

Subtraction

```
tf.subtract(20,10)
```

### Multiplication

```
tf.multiply(10,5)
```

### Division

```
tf.divide(20,5)
```

### Square

```
tf.square(5)
```

### Square Root

```
tf.sqrt(25.0)
```

## Tensor Shape

```
tensor.shape
```

## Reshape Tensor

```
tf.reshape(tensor, (2,2))
```

## Convert NumPy Array to Tensor

```
import numpy as np

arr = np.array([1,2,3])

tensor = tf.convert_to_tensor(arr)
```

## Neural Network using Keras

```
from tensorflow import keras model = keras.Sequential([ keras.layers.Dense(64, a
```

## Compile Model

```
Compile Model
model.compile(
    optimizer="adam",
    loss="mse",
    metrics=["accuracy"]
)
```

## Train Model

```
model.fit( X_train, y_train, epochs=10 )
```

## Evaluate Model

```
model.evaluate( X_test, y_test )
```

## Prediction

```
model.predict(X_test)
```

## Saving Model

```
model.save("model.h5")
```

## Loading Model

```
from tensorflow.keras.models import load_model

model = load_model("model.h5")
```

## TensorFlow Modules

tensorflow.keras tensorflow.data tensorflow.image tensorflow.math tensorflow.random  
tensorflow.nn tensorflow.linalg tensorflow.io

## Keras

Keras is a high-level API of TensorFlow used to build deep learning models easily.

## Example

```
from tensorflow import keras
```

## Advantages

- Easy to use
- Less coding
- Fast model development

## Activation Functions

- ReLU
- Sigmoid
- Softmax
- Tanh
- Linear

## Optimizers

Double-click (or enter) to edit

- Adam
- SGD
- RMSprop
- Adagrad
- Adamax

## Loss Functions

-Mean Squared Error (MSE)- -cBinary Crossentropy

- Categorical Crossentropy
- Sparse Categorical - - - Crossentropy

## Common Layers

- Dense
- Conv2D
- MaxPooling2D
- Flatten
- Dropout
- LSTM -Embedding

## TensorFlow Applications

- Machine Learning
- Deep Learning
- Artificial Intelligence
- Image Recognition
- Object Detection -Face Recognition
- Speech Recognition
- Natural Language Processing (NLP)
- Chatbots

## Tensor flow logo



## ✓ TensorFlow vs Scikit-learn

| TensorFlow                                                          | Scikit-learn                                                 |
|---------------------------------------------------------------------|--------------------------------------------------------------|
| Used for <b>Deep Learning</b>                                       | Used for <b>Machine Learning</b>                             |
| Supports <b>Neural Networks</b>                                     | Supports <b>Traditional Machine Learning Algorithms</b>      |
| Supports <b>CPU, GPU, and TPU</b>                                   | Mostly uses <b>CPU</b>                                       |
| Best for <b>Large Datasets</b>                                      | Best for <b>Small and Medium-sized Datasets</b>              |
| More <b>Complex</b> to learn                                        | <b>Easy</b> to learn and use                                 |
| Developed by <b>Google</b>                                          | Developed by the <b>Scikit-learn Community</b>               |
| Suitable for <b>Deep Learning and AI</b> applications               | Suitable for <b>Traditional ML</b> tasks like classification |
| Uses <b>Keras API</b> for building models                           | Provides built-in ML algorithms without deep learning        |
| Used in <b>Image Recognition, NLP, Speech Recognition, Robotics</b> | Used in <b>Spam Detection, Recommendation Systems</b>        |
| Better for <b>Production-level AI Systems</b>                       | Better for <b>Learning, Research, and Traditional ML</b>     |

### Future of TensorFlow

- Better support for Artificial Intelligence and Deep Learning.
- Faster training using advanced GPUs and TPUs.
- More automation in model development.
- Improved cloud and edge device deployment.
- Better support for large-scale data processing.
- Growth in healthcare, finance, robotics, and autonomous vehicles.
- Easier integration with AI development tools. Continuous improvements from Google's developer community.
- More efficient and optimized machine learning models. Increased adoption in research and industry.

